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Research paper

Decoding depression with computer vision-assisted analysis of synchronized facial expressions

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Highlights

- Self-reports are widely used but are prone to bias, underscoring the need for objective behavioral markers.
- We used naturalistic paradigm and ISC to find diagnostic cues for depression in facial expression synchronization patterns.
- Classification models trained with facial expression synchronization successfully detected depression at 72–90% accuracy.
- Our findings show diagnostic information for depression is encoded within dynamic facial expression patterns.

Abstract

Background

Facial expressions convey nonverbal signals of internal emotional states and hold potential as important markers for affective disorders such as depression.

Methods

Here, we adopted a naturalistic paradigm and combined with an inter-subject correlation (ISC) framework to identify diagnostic cues for depression from facial expression synchronization patterns while naturally responding to emotionally charged videos. Using timeseries activity patterns of action units (AUs) extracted from facial expressions of emotion, we developed models for detecting depression with AU-ISC vector as key features.

Results

Resulting models successfully detected depression at 72–90% accuracy. In contrast, models using mean AU activity exhibited poor performance, showcasing the diagnostic utility of the AU-ISC models.

Conclusions

These findings underscore the potential utility of facial expression data in advancing objective and clinically meaningful diagnostic tools for depression. More broadly, the present study highlights a new direction for emotion research, utilizing facial expression dynamics as a complementary tool alongside established self-report measures.

Introduction

Facial expressions serve as a key channel through which internal emotional states are externally conveyed and communicated to others (Andrews, 1963; Darwin, 1999; Ekman, 1999). Studying facial expressions has traditionally provided valuable insights into the mechanisms underlying emotional experience and expression (Dols and Russell, 2017), complementing findings based on self-report (Izard, 2007). While self-report measures are among the most commonly used methods for assessing emotions (Anvari et al., 2022; Larsen and Prizmic-Larsen, 2006), they are subject to various biases, including socially desirable responding, acquiescent responding, and extreme responding (Paulhus and Vazire, 2007). These biases can compromise the accuracy of self-reported emotional experiences, making objective emotion assessment more challenging. Facial expressions, in contrast, may be able to provide more objective measure that is less susceptible to such biases.

A widely used approach for analyzing facial expressions is Facial Action Coding System (FACS; Ekman and Friesen, 1978), which was developed using the anatomical basis of facial movements and describes facial movements in terms of Action Units (AUs). Historically, analysis of facial expressions relied on manual coding by certified FACS experts, a method that was not only time-consuming and labor-intensive but often impractical for large-scale or real-time applications. Recent development of open-source tools for AU analysis, such as Python Facial Expression Analysis Toolbox (Py-Feat; Cheong et al., 2023) and OpenFace (Baltrušaitis et al., 2018), allowed for machine learning-based automated AU analysis. Automated facial expression analysis has broad applications in many fields, encompassing emotion and paralinguistic communication, clinical psychology, neurology, and beyond (Tian et al., 2005). In clinical settings where diagnoses largely depend on patients' self-reports, identifying objective behavioral markers—such as facial expressions—using automated AU analysis could enhance the accuracy of psychiatric diagnoses and treatments. Indeed, recent studies explored the use of audio features during speech (Ye et al., 2021), neural and behavioral parameters (Liu et al., 2025), and facial expression data from clinical patients (Ye et al., 2025) to improve diagnosis and treatment in clinical settings.

Major depressive disorder (MDD) is a prevalent mood disorder, characterized by a sustained feelings of sadness or a lack of interest or pleasure in most activities (American Psychiatric Association, 2013). Given its strong affective nature, a systematic investigation of emotion expression is warranted (Rottenberg and Vaughan, 2008). Among its affective symptoms, anhedonia—marked reduction in interest or enjoyment of typically pleasurable experiences—is a crucial feature that distinguishes depression from other comorbid affective disorders (Nitschke et al., 2001; Watson et al., 1995), and its strong association with depression has been demonstrated through self-report measures (Davidson, 1998). Indeed, anhedonic symptoms in individuals with depression have also been observed as reduced positive expressive behaviors, such as the frequency and duration of smiling, eyebrow movements, head movements, and gaze compared to control participants (Ellgring, 1989; Fossi et al., 1984; Gotlib and Robinson, 1982; Jones and Pansa, 1979; Troisi and Moles, 1999). Additionally, in studies where participants were exposed to positive emotion-inducing situations (Berenbaum and Oltmanns, 1992) or asked to view positive emotionally charged stimuli (Sloan et al., 2001), individuals with depression displayed fewer and less intense positive facial expressions than the controls.

Building on these findings, the present study aims to analyze and compare patterns of facial expressions between depressed and healthy individuals. Of note, we sought to test the potential diagnostic utility of facial expressions by leveraging the inter-subject correlation (ISC) framework (Hasson et al., 2004). ISC is often used to examine the degree of neural synchronization across individuals while they experienced the same time-locked stimuli using functional neuroimaging data (Nastase et al., 2019), which can be extended to other types of timeseries data (e.g., temporal pattern of FACS-based AU activity) (Chou et al., 2025). Unlike previous research, which mostly focused on comparing the frequency or duration of emotional expression behaviors between groups (Girard et al., 2013; Kim et al., 2023), ISC analysis assesses how similar the temporal dynamics of facial expressions are within each group. Such analysis allows for continuous, fine-grained tracking of emotional dynamics without interrupting the natural flow of emotional responses. This makes it particularly useful for capturing spontaneous emotional reactions in real-world settings. By integrating this approach with FACS, we can analyze the temporal dynamics of individual AUs, allowing for a more precise identification of facial regions that provide key diagnostic cues for distinguishing depressed individuals from healthy controls. Importantly, this approach

aligns with the pathological feature of depression, heterogeneity in symptom manifestation across individuals (Goldberg, 2011). Such variability can be reflected in the temporal coordination—or lack thereof—of emotional expressions within a group. Whereas mean AU activation may overlook these subtle temporal divergences, AU-ISC can capture the degree of synchrony or deviation across individuals, providing a more sensitive marker of depression-related differences in emotional expression dynamics.

To leverage the temporal dynamics of emotional facial expressions, we adopted a naturalistic paradigm wherein videos designed to elicit positive and negative emotions were shown to participants. Naturalistic paradigms are defined as experimental designs that employ rich, multimodal dynamic stimuli that represent daily experience, such as film clips, TV advertisements, news items, and spoken narratives (Sonkusare et al., 2019). In cognitive neuroscience, this paradigm is considered a solution to the limitations of traditional, highly controlled laboratory experiments with limited ecological validity (Finn et al., 2020). Furthermore, it was suggested that carefully selected films can effectively elicit discrete emotions, and that comparable neutral stimuli can control the effects of simply watching a film (Gross and Levenson, 1995). Thus, this approach would capture more natural and organic facial expressions.

This present study aimed to investigate the differences in facial expressions between depressed and healthy individuals and explore their potential diagnostic value. Specifically, we examined the extent to which participants exhibit similar temporal patterns of facial expressions while watching emotionally charged videos. To address these questions, we first tested whether participants within each group display synchronized facial expressions in response to the same emotionally charged videos. We expected healthy individuals to show a higher degree of facial expression synchrony during positive emotional experience, as depression is associated with reduced emotional expression in response to positive stimuli (Berenbaum and Oltmanns, 1992; Sloan et al., 2001). We also examined group differences in facial expression activation by comparing the mean intensity of facial expressions. Finally, using a machine learning-based algorithm, we trained a classifier using only facial expression synchrony patterns and tested whether it could reliably distinguish between the two groups. If so, these patterns could serve as potential behavioral markers of depression.

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Participants

Ninety-eight participants (67 females; age range=19–34years, *M*=21.49, *SD*=3.14) were recruited between May and November of 2024. Prior to the experiment, participants completed the Patient Health Questionnaire-9 (PHQ-9) (Spitzer et al., 1999). Based on the PHQ-9 scores, participants were assigned to either the Healthy Control (HC) group or the Depressed (DP) group for an in-person clinical interview. Individuals with a PHQ-9 score above 10 were considered potential candidates for the DP …

Whole-face ISC

ISCs of each AU were computed separately for DP and HC groups. For *Happy 1*, the DP group exhibited the highest ISC in AU14 (ISC=0.12, $q<0.001$). In contrast, the HC group showed high ISCs in AU06 (ISC=0.21, $q<0.001$), AU12 (ISC=0.19, $q<0.001$), and AU14 (ISC=0.19, $q<0.001$). For *Happy 2*, in the DP group, AU06 (ISC=0.32, $q<0.001$), AU09 (ISC=0.22, $q<0.001$), AU12 (ISC=0.30, $q<0.001$), and AU14 (ISC=0.29, $q<0.001$) showed high ISCs. Similarly, in the HC group, AU06 …

Discussion

Here, we used a naturalistic viewing paradigm to identify and compare synchronized facial expressions in response to emotionally charged videos between depressed and healthy individuals. Our results indicated that a machine learning

classifier trained using ISC of AU timeseries data reliably detected depression at 72–90% accuracy. AU-ISC models trained on either positively or negatively charged videos showed comparable classification performance. In contrast, mean AU activity or emotion scores …

CRediT authorship contribution statement

Seohyeon Lee: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yunsu Kim:** Writing – review & editing, Methodology, Investigation. **Hayoung Ryu:** Writing – review & editing, Methodology, Investigation. **Sunkyung Yoon:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **M. Justin Kim:** Writing – review & editing, Writing – original draft, …

Code availability

Codes for analyses and visualizations are accessbile at https://doi.org/10.5281/zenodo.17596629 ↗. …

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Declaration of competing interest

The authors declare no conflict of interest. …

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